

Zhihui Tian

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Summary

- Ph.D. candidate in Electrical and Computer Engineering at the University of Florida developing scalable machine learning methods for large-scale 3D microstructure evolution. Combines graph neural networks and physics-regularized modeling to improve the efficiency and long-term stability of mesoscale microstructure evolution prediction. Seeking industry research / ML roles and postdoctoral positions.

Education

- Ph.D. University of Florida**, Electrical and Computer Engineering Gainesville, FL
May 2023 – present
- Advised by Prof. Joel B. Harley (SmartDATA Lab).
 - Research on scalable, physics-guided machine learning for 3D microstructure evolution.
- M.S. University of Florida**, Electrical and Computer Engineering Gainesville, FL
Aug 2021 – May 2023

Experience

- Lawrence Livermore National Laboratory (LLNL)**, Collaborative Research Project Remote / Gainesville, FL
Sept 2025 – present
1 year 1 month
- Scalable hybrid CNN-GNN surrogate for 3D grain growth.
- Conducted computational experiments and benchmarking for a hybrid bijective autoencoder and graph neural network (AE+GNN) surrogate model.
 - Developed the latent-space inference strategy and conducted scalability studies, achieving up to 117x memory reduction and 115x runtime improvement on 3D meshes up to 160^3 relative to a GNN-only baseline.
 - Established and applied quantitative statistical benchmarks to evaluate surrogate predictions against Monte Carlo ground truth.
- SmartDATA Lab, University of Florida**, Graduate Research Assistant Gainesville, FL
May 2023 – present
3 years 5 months
- Physics-regulated interpretable ML for 3D microstructure evolution: developed a scalable 3D architecture with strong spatiotemporal extrapolation across large simulation domains; extended the physics-regulated framework to inclination- and misorientation-dependent grain boundary dynamics; applied the model to Lab-DCT experimental data with quantitative comparison against ML predictions.
 - Computer vision and deep learning for experimental microstructure analysis: developed and optimized a TEM image preprocessing pipeline that improves registration accuracy; adapted and fine-tuned U-Net and the Segment Anything Model for microstructure segmentation of registered TEM images.
 - Gaussian-sampled mode filter for learning and emulating microstructure evolution: designed Gaussian-sampled kernels that emulate grain boundary motion and mitigate lattice pinning artifacts; demonstrated improved von Neumann-Mullins law adherence over classical Monte Carlo Potts models; analyzed energy-mobility interactions for interpretable modeling.

Publications

- 3D-PRIMME: A scalable physics-guided machine learning framework for 3D grain growth**
Zhihui Tian, Kang Yang, Vishal Yadav, Amanda R. Krause, Michael R. Tonks, Joel B. Harley
arxiv.org/abs/2607.04680 (Materials & Design, 2026)

Scaling kinetic Monte-Carlo simulations of grain growth with combined convolutional and graph neural networks

Zhihui Tian, Ethan Suwandi, Tomas Oppelstrup, Vasily V. Bulatov, Joel B. Harley, Fei Zhou
doi.org/10.1016/j.actamat.2026.122153 (Acta Materialia, 2026)

Beyond curvature-driven grain growth: Insights from fully anisotropic Monte Carlo Potts simulations

Lin Yang, Vivekanand Muralikrishnan, Vishal Yadav, Zhihui Tian, Joel B. Harley, Amanda Krause, Michael Tonks
doi.org/10.1016/j.actamat.2025.121672 (Acta Materialia, 2026)

Improved PCA reconstruction-based unsupervised anomaly detection in uncontrolled structural health monitoring with correntropy

Kang Yang, Zhenhan Lin, Zekun Yang, Zhihui Tian, Jie Ma, Jose C. Principe, Joel B. Harley
ieeexplore.ieee.org/document/11078390 (IEEE Transactions on Industrial Informatics, 2025)

Baseline optimized autoencoder-based unsupervised anomaly detection in uncontrolled dynamic structural health monitoring

Kang Yang, Tianqi Liu, Zekun Yang, Yang Zhou, Zhihui Tian, Nam H. Kim, Joel B. Harley
doi.org/10.1177/14759217251324107 (Structural Health Monitoring, 2025)

Quantifying heterogeneous ecosystem services with multi-label soft classification

Zhihui Tian, John Upchurch, G. Austin Simon, Jose Dubeux, Alina Zare, Chang Zhao, Joel B. Harley
doi.org/10.1109/IGARSS53475.2024.10642804 (IEEE International Geoscience and Remote Sensing Symposium (IGARSS), 2024)

Weight decay optimized unsupervised autoencoder-based anomaly detection in uncontrolled dynamic structural health monitoring

Kang Yang, Zekun Yang, Zhihui Tian, Joel B. Harley
doi.org/10.1007/978-3-031-94895-4_3 (Dynamic Data Driven Applications Systems (DDDAS), 2024)

Skills

Machine Learning

Graph neural networks, convolutional autoencoders, physics-informed / physics-regularized models, surrogate modeling, unsupervised anomaly detection

Computer Vision

Image segmentation (U-Net, Segment Anything Model), image registration and preprocessing, TEM and Lab-DCT microstructure data

Materials Simulation

Grain growth, kinetic Monte Carlo Potts models, mode filter models, 3D microstructure evolution, von Neumann-Mullins analysis

Programming and Tools

Python, PyTorch, PyTorch Geometric, NumPy / SciPy, MATLAB, Git, Linux, HPC clusters

Languages

Chinese (Mandarin)

Native speaker

English

Fluent